

## Instructions

### Q1 – UNIX and Terminal (20 points, individual, CLO 1)

Every member does this part **on their own Mac account** and includes their own screenshots in the documentation. Use the **Terminal** application only (no Finder for a–c).

- **a) (6 points)** Create the complete project folder tree below with `mkdir` and `touch` (use `mkdir -p` for nested folders). Take a screenshot of the Terminal showing the commands and the output of `ls -R` from inside `sitename_team#`.

```
sitename_team#/          (e.g. codecraft_team3/)
├── index.html             (home page)
├── about.html             (team page)
├── lesson_firstname.html  (one per team member, e.g. lesson_sara.html)
├── quiz.html
├── references.html
├── assets/
│   ├── css/
│   │   └── style.css      (the ONLY stylesheet of the site)
│   └── media/             (all images, video and audio files)
```

```
josh@Josh-MacBook-Air ~ % cd Desktop
josh@Josh-MacBook-Air Desktop % pwd
/Users/josh/Desktop
josh@Josh-MacBook-Air Desktop % mkdir -p team5site_team5/assets/css team5site_team5/assets/media
josh@Josh-MacBook-Air Desktop % cd team5site_team5
josh@Josh-MacBook-Air team5site_team5 % touch index.html about.html quiz.html references.html assets/css/style.css
josh@Josh-MacBook-Air team5site_team5 % touch lesson_nick.html lesson_fadil.html lesson_zainab.html
josh@Josh-MacBook-Air team5site_team5 % ls -R
about.html          lesson_nick.html
assets              lesson_zainab.html
index.html          quiz.html
lesson_fadil.html   references.html
lesson_josh.html

./assets:
css      media

./assets/css:
style.css

./assets/media:
josh@Josh-MacBook-Air team5site_team5 %
```

- **b) (3 points)** Create a temporary file `test.html`, rename it to `draft.html` with `mv`, then delete it with `rm`. Copy `index.html` to `backup_index.html` with `cp`, then delete the copy. Screenshot the commands and an `ls -l` after each step.

```
josh@Josh-MacBook-Air team5site_team5 % touch test.html
ls -l test.html
mv test.html draft.html
ls -l draft.html
rm draft.html
cp index.html backup_index.html
ls -l backup_index.html
rm backup_index.html
-rw-r--r--  1 josh  staff   0 26 Sep 17:04 test.html
-rw-r--r--  1 josh  staff   0 26 Sep 17:04 draft.html
-rw-r--r--  1 josh  staff   0 26 Sep 17:04 backup_index.html
```

- **c) (5 points)** Set permissions so that folders are 755 and files are 644. Show **both** notations: once with numeric `chmod` and once with symbolic `chmod` (`u=rwx,g=rx,o=rx` / `u=rw,g=r,o=r`). Screenshot `ls -l` of the project folder and of `assets/media`.

```
josh@Josh-MacBook-Air team5site_team5 % chmod 755 . assets assets/css assets/media
chmod 644 *.html assets/css/style.css
ls -l
ls -l assets
total 0
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 about.html
drwxr-xr-x  4 josh  staff 128 26 Sep 16:54 assets
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 index.html
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 lesson_fadil.html
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 lesson_josh.html
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 lesson_nick.html
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 lesson_zainab.html
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 quiz.html
-rw-r--r--  1 josh  staff    0 26 Sep 16:59 references.html
total 0
drwxr-xr-x  3 josh  staff  96 26 Sep 16:59 css
drwxr-xr-x  2 josh  staff  64 26 Sep 16:54 media
```

- **d) (6 points)** Written answers (in the documentation file):
  - From assets/css/style.css, what is the **relative** path to assets/media/logo.png?
    - ➔ ../media/logo.png
  - What is the relative path from index.html to style.css? What is the **absolute** path of style.css on the RIT server if your project is in /home/username/public\_html/?
    - ➔ assets/css/style.css
    - ➔ /home/jt2737/www/behindthescreen\_team5/assets/css/style.css
  - Explain what 755 and 644 mean (owner / group / others). Why should HTML files not be executable?
    - ➔ 755 (rwxr-xr-x): Basically with this you can, read, write, and execute... others have read and execute. We use it for directories so the web server can enter and list contents.
    - 644 (rw-r--r--): This is just read and write. Others have read only. We use it for static files
    - HTML files are static text documents that are rendered by client-side web browsers, not executable programs or server scripts. If we gave those files executable permissions it is unnecessary security vulnerabilities since it's not supposed to do much.
  - Give the single command that would set quiz.html to rw-r----- and explain what a visitor of your website would see afterwards and why.
    - ➔ chmod 640 quiz.html ... visitor sees a **403 Forbidden** error page. Setting permissions to 640 removes read access for others.